Sina Abdipoor (思纳), Ph.D.

AI Engineer specializing in Optimization, Operations Research, and Machine Learning

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Summary

Operations Research and AI engineer with a Ph.D. in Computer Science and more than 5 years of experience building optimization and machine learning solutions for scheduling, timetabling, and resource management. I turn complex operational problems into models that save time and money and enable data-driven decisions.

Experience

Sunway University, Lecturer, Department of Data Science and Artificial Intelligence — Dec 2025 – present

- Lead ScholORs, an optimization research group working on real-world scheduling and timetabling, with 9 papers in 2026.
- Teach optimization, applied machine learning, and software engineering, and help design the Master in Artificial Intelligence curriculum.
- Supervise 2 Master's students and more than 20 undergraduate final year projects.

Universiti Putra Malaysia, Graduate Researcher & Tutor — Sep 2020 – Sep 2025

- Competed in the International Timetabling Competition 2019 with hybrid metaheuristics for large university timetabling problems with student sectioning.
- Turned the results into papers in IEEE Access and Intelligent Systems with Applications, and presented them at international conferences.
- Taught Computer Programming 2 and Machine Learning to undergraduate and postgraduate students.

Independent Consultant, Operations Research Engineer — Jan 2020 – Mar 2025

- Built and deployed scheduling, timetabling, and resource management models that helped companies save time and money.
- Analyzed client workflows to find where time and resources were lost.
- Turned model results into clear recommendations that enabled data-driven decisions by management.

Pars Six Sigma Excellence, Software Engineer — Jul 2017 – Jul 2019

- Designed, built, and maintained a CRM system for employees and customers in a small software team.
- Developed the Java backend with a variety of libraries, frameworks, and APIs, and designed the SQL database.

IKCO, Android Developer (Internship) — Jun 2016 – Sep 2016

Built an Android app that lets assembly line workers check parts stock and order parts from their station.

Skills

Operations Research: Mixed Integer Programming, Constraint Programming, Stochastic Optimization, Metaheuristics (Simulated Annealing, Genetic Algorithms, Local Search, Hybrid Algorithms), Scheduling and Timetabling, Gurobi, Google OR-Tools

Machine Learning: Predictive Modeling, Time Series Forecasting, Deep Learning (RNN, LSTM, Graph Neural Networks), Reinforcement Learning, TensorFlow, Scikit-Learn, Pandas, NumPy

Programming and Tools: Python, Java, SQL, Git, Power BI, Tableau, Excel

Languages: English (Bilingual, IELTS 8), Mandarin (Intermediate), Malay (Basic)

Projects

ESA Space Optimisation Competition (SpOC 4) — 2026

Ranked 3rd globally in the Luna Tomato Advertising challenge of the European Space Agency's space logistics optimization competition at GECCO 2026, using permutation encodings and simulated annealing. Our solutions were published as two GECCO 2026 papers.

International Timetabling Competition 2019 2022
Took part in the International Timetabling Competition 2019, the latest benchmark for university course timetabling, with my own implementation of the benchmark. It became the core of my Ph.D. and led to several papers and conference talks.

Java Framework for Genetic Algorithms 2019
A Java framework to implement evolutionary algorithms, apply them to different problems, and compare their performance. Built during my master's research.

Auto Parts Finder 2016
An Android app used by assembly line workers at IKCO to check parts stock in real time and order parts, which cut idle time on the line.

Education

Ph.D. **Universiti Putra Malaysia** 2020 – 2026
Computer Science
Thesis: Optimality and Generality of Metaheuristics in University Course Timetabling. CGPA: 4.0/4.0

M.Sc. **K. N. Toosi University of Technology** 2016 – 2019
Computer Engineering, Artificial Intelligence and Robotics
Thesis: Analyzing Parameter Tuning of Evolutionary Algorithms in Optimization Problems

Selected Publications

An integration of discrete wavelet transform and time-series recurrent neural network to improve stock price prediction Aug 2026
Applied Computing and Informatics

Optimality Versus Generality: Performance Assessment of Meta-Heuristics in Educational Timetabling 2025
IEEE Access

Hybrid Graph Neural Networks with LSTM Attention Mechanism for Recommendation Systems in MOOCs Mar 2024
2024 20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)
Best Paper Award

Awards

Best Paper Award Aug 2026
19th International Conference on Advanced Computer Theory and Engineering (ICACTE)

National Finalist, 3 Minute Thesis (3MT) Competition Jun 2024
Universiti Kebangsaan Malaysia

Winner, 3 Minute Thesis (3MT) Competition Mar 2024
Universiti Putra Malaysia

Talks

Applied Operations Research in Driving Sustainability China
Invited talks, Baoshan University and Hefei University Aug 2026

Applied Operations Research in Developing Sustainable Cities Malaysia
Organizer and speaker, Sunway Tech Club, Sunway University Aug 2026

Certificates

Chinese (Mandarin) Certificate May 2021
Peking University

Google Data Analytics Certificate Apr 2021
Google